

Decent Coaching Classes

VIII CBSE

Mind Your Math - IV

Answer Key — Algebraic Expressions and Polynomials

Class VIII — Mathematics (NCERT) | Maximum Marks: 30

Section A — Multiple Choice Questions

A1. (d) -5 [1]

A2. (c) trinomial [1]

A3. (a) $3a$ [1]

A4. (b) $5x^2y$ and $-7x^2y$ [1]

A5. (c) 4 terms [1]

A6. (b) 4 [1]

Section B — Fill in the Blanks / Very Short Answer

A7. -3 [1]

A8. like [1]

A9. monomial [1]

A10. three [1]

Section C — Short Answer Questions

A11. (a) $a^2 + a - 5$ has three unlike terms, so it is a Trinomial. (b) $7ab - 4$ has two unlike terms, so it is a Binomial. [2]

A12. (a) $4x^3 - 10x^{-1} + 1$ is NOT a polynomial because the term $10x^{-1}$ has a negative power of the variable x ; in a polynomial, every variable must have a whole-number (non-negative integer) exponent. (b) $5x - 2y + 7$ IS a polynomial (a trinomial), since every variable has a non-negative integer power. [2]

A13. $(3a - 2ab - 6) - (4a + 5ab - 7) = 3a - 2ab - 6 - 4a - 5ab + 7 = (3a - 4a) + (-2ab - 5ab) + (-6 + 7) = -a - 7ab + 1$ [2]

A14. $(3a + 5b)(5a + 7b) = 15a^2 + 21ab + 25ab + 35b^2 = 15a^2 + 46ab + 35b^2$ [2]

A15. Grouping like terms: $(6a - 2a) + (13b^2 - b^2) + 2a^2 + bc = 4a + 12b^2 + 2a^2 + bc$, i.e. $2a^2 + 4a + 12b^2 + bc$ [2]

A16. $2x(x^2 - y - z) = 2x \cdot x^2 - 2x \cdot y - 2x \cdot z = 2x^3 - 2xy - 2xz$ [2]

Section D — Long Answer Questions

A17. $(x + y)(x^2 - xy + y^2) = x \cdot x^2 - x \cdot xy + x \cdot y^2 + y \cdot x^2 - y \cdot xy + y \cdot y^2 = x^3 - x^2y + xy^2 + x^2y - xy^2 + y^3$. The terms $-x^2y$ and $+x^2y$ cancel, and $+xy^2$ and $-xy^2$ cancel, leaving: $x^3 + y^3$ [4]

A18. First expand $(2x + 7y)(x - 3y) = 2x^2 - 6xy + 7xy - 21y^2 = 2x^2 + xy - 21y^2$. Next expand $(9x + 4)(3x - 2) = 27x^2 - 18x + 12x - 8 = 27x^2 - 6x - 8$. Now subtract: $(2x^2 + xy - 21y^2) - (27x^2 - 6x - 8) = 2x^2 + xy - 21y^2 - 27x^2 + 6x + 8 = -25x^2 + xy - 21y^2 + 6x + 8$ [4]

— *End of Answer Key* —